

SparkBACnet

SparkBACnet is a BACnet/IP client driver for building automation systems.

Version 0.1.1 Ignition 8.1.33+ and 8.3 Runs on: any gateway that reaches the BACnet/IP network

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1. Overview

SparkBACnet is a standalone device driver, licensed and running independently of any other SparkBridge module. Module id `com.greenguardusa.sparkbacnet`.

2. What You Need

- Ignition 8.1.33 or newer, or the 8.3 line (separate build).
- A BACnet/IP device reachable over UDP, port 47808.

3. Install

Install the signed module through Config > Modules > Install or Upgrade a Module; no build step is required. Module id `com.greenguardusa.sparkbacnet`.

4. License

Licensed per gateway, bought once and kept, soft-licensed, entitlement `com.greenguardusa.sparkbacnet`.

From release 2.7.0, every SparkBridge module runs on Ignition's standard two-hour module trial (the gateway's Reset Trial button restarts the clock). A verified key overrides the trial entirely and the trial clock is then ignored. Drop the key at `<data>/sparkbridge-license.key`; each module re-checks the file every 30 seconds, so installing, replacing, or removing a key takes effect with no restart.

5. Configure

Config > OPC UA > Device Connections > Create new Device > the BACnet device type. Discovery uses Who-Is/I-Am. Addressing grammar: `dev:1234/AT:1.PV` (device instance, object type and instance, property). Supported services include ReadProperty, ReadPropertyMultiple, WriteProperty (priority array 1-16, relinquish via NULL), and SubscribeCOV with lifetime renewal. Supported object types: analog/binary/multistate input/output/value, device, and read-only schedule/calendar.

Firewall notes. BACnet/IP uses UDP 47808 (BVLN/NPDU/APDU); ensure the gateway's UDP port is reachable to and from field devices, including broadcast for Who-Is discovery.

6. Operate

Writes use the standard BACnet priority array (1-16); relinquish a write with a NULL value rather than leaving a stale priority slot occupied. SubscribeCOV renews on its lifetime; watch for subscriptions lapsing on devices that reboot or drop network.

7. Verify

Wire encoding has been verified against a real bacnet-stack server: Who-Is/I-Am, ReadProperty, ReadPropertyMultiple, and WriteProperty with priority and relinquish all passed. SubscribeCOV notification receipt was not exercised in that run. A configured Ignition device against a real controller, browsing and reading through OPC UA, and 8.3 support are tracked separately; see the README in the bundle for current status before deploying.

8. Troubleshoot

Symptom	Cause / fix
No devices found on discovery	Confirm UDP 47808 broadcast reaches the field network from the gateway; a router or VLAN boundary commonly blocks BACnet broadcast.
COV subscription stops delivering	SubscribeCOV notification receipt has known open items per the README; verify current status there before relying on it in production.
Write does not take effect	Check the priority array; a higher-priority writer elsewhere on the network can override a lower-priority write.

9. Reference

Module id	com.greenguardusa.sparkbacnet
Default port	UDP 47808 (BACnet/IP)
Addressing grammar	dev:<instance>/<ObjectType>:<instance>.<property>
License entitlement	com.greenguardusa.sparkbacnet

10. Support

Docs page section: </sparkbridge/docs#sparkbacnet>. Contact: </sparkbridge/contact>. Release downloads and checksums: [GitHub releases](#).